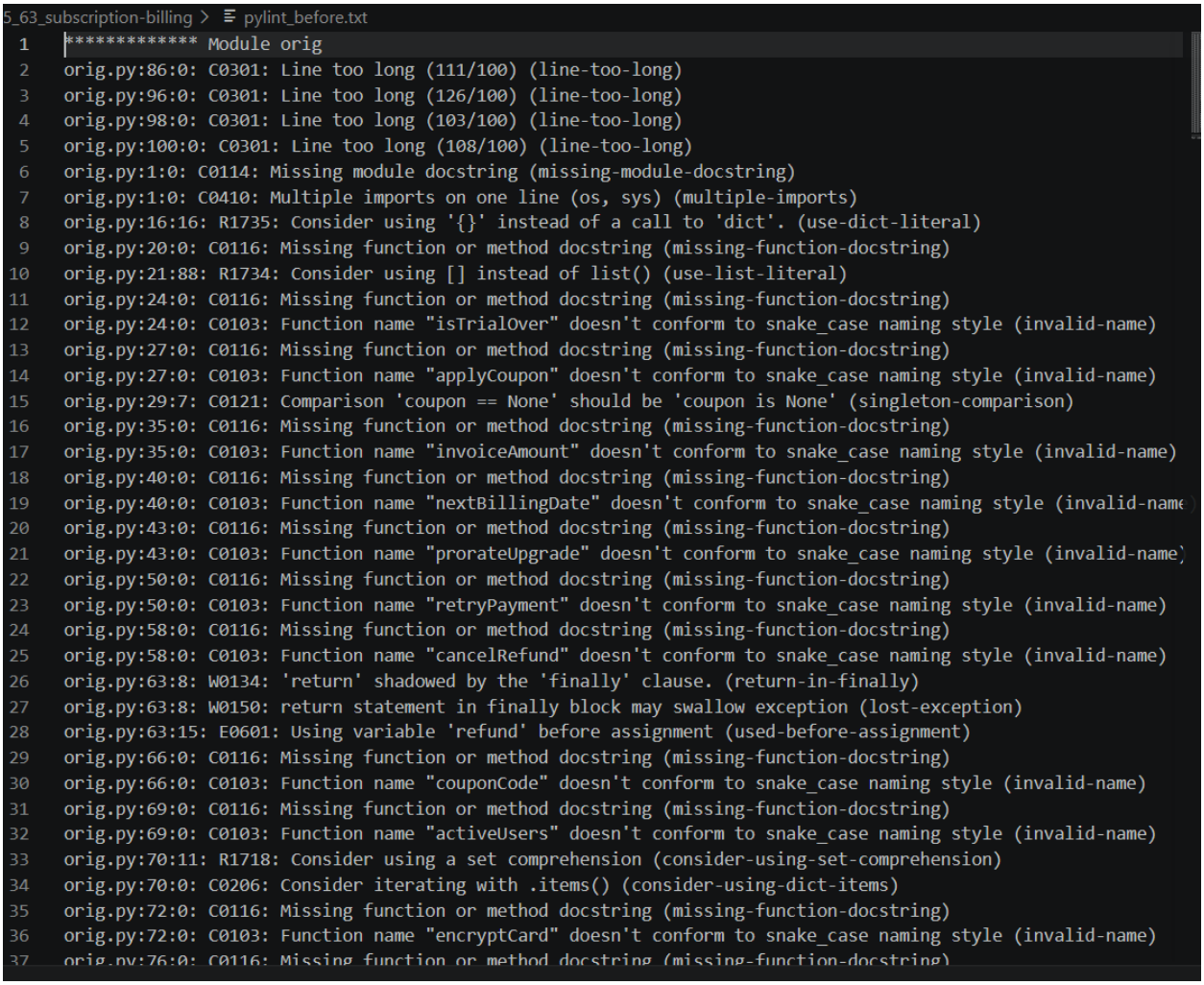
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LAB -05	SUB: SEP

Lab 5: Static Code Analysis Report

Program: billing_system.py (Subscription Billing System)

BEFORE PLAYINT



5_63_subscription-billing > pylint_before.txt

```
1  ************* Module orig
2  orig.py:86:0: C0301: Line too long (111/100) (line-too-long)
3  orig.py:96:0: C0301: Line too long (126/100) (line-too-long)
4  orig.py:98:0: C0301: Line too long (103/100) (line-too-long)
5  orig.py:100:0: C0301: Line too long (108/100) (line-too-long)
6  orig.py:1:0: C0114: Missing module docstring (missing-module-docstring)
7  orig.py:1:0: C0410: Multiple imports on one line (os, sys) (multiple-imports)
8  orig.py:16:16: R1735: Consider using '{}' instead of a call to 'dict'. (use-dict-literal)
9  orig.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
10 orig.py:21:88: R1734: Consider using [] instead of list() (use-list-literal)
11 orig.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
12 orig.py:24:0: C0103: Function name "isTrialOver" doesn't conform to snake_case naming style (invalid-name)
13 orig.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
14 orig.py:27:0: C0103: Function name "applyCoupon" doesn't conform to snake_case naming style (invalid-name)
15 orig.py:29:7: C0121: Comparison 'coupon == None' should be 'coupon is None' (singleton-comparison)
16 orig.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
17 orig.py:35:0: C0103: Function name "invoiceAmount" doesn't conform to snake_case naming style (invalid-name)
18 orig.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
19 orig.py:40:0: C0103: Function name "nextBillingDate" doesn't conform to snake_case naming style (invalid-name
20 orig.py:43:0: C0116: Missing function or method docstring (missing-function-docstring)
21 orig.py:43:0: C0103: Function name "prorateUpgrade" doesn't conform to snake_case naming style (invalid-name)
22 orig.py:50:0: C0116: Missing function or method docstring (missing-function-docstring)
23 orig.py:50:0: C0103: Function name "retryPayment" doesn't conform to snake_case naming style (invalid-name)
24 orig.py:58:0: C0116: Missing function or method docstring (missing-function-docstring)
25 orig.py:58:0: C0103: Function name "cancelRefund" doesn't conform to snake_case naming style (invalid-name)
26 orig.py:63:8: W0134: 'return' shadowed by the 'finally' clause. (return-in-finally)
27 orig.py:63:8: W0150: return statement in finally block may swallow exception (lost-exception)
28 orig.py:63:15: E0601: Using variable 'refund' before assignment (used-before-assignment)
29 orig.py:66:0: C0116: Missing function or method docstring (missing-function-docstring)
30 orig.py:66:0: C0103: Function name "couponCode" doesn't conform to snake_case naming style (invalid-name)
31 orig.py:69:0: C0116: Missing function or method docstring (missing-function-docstring)
32 orig.py:69:0: C0103: Function name "activeUsers" doesn't conform to snake_case naming style (invalid-name)
33 orig.py:70:11: R1718: Consider using a set comprehension (consider-using-set-comprehension)
34 orig.py:70:0: C0206: Consider iterating with .items() (consider-using-dict-items)
35 orig.py:72:0: C0116: Missing function or method docstring (missing-function-docstring)
36 orig.py:72:0: C0103: Function name "encryptCard" doesn't conform to snake_case naming style (invalid-name)
37 orig.py:76:0: C0116: Missing function or method docstring (missing-function-docstring)
```

```
orig.py:84:0: C0116: Missing function or method docstring (missing-function-docstring)
orig.py:84:0: C0103: Function name "renderInvoice" doesn't conform to snake_case naming style (invalid-name)
orig.py:86:20: W1406: The u prefix for strings is no longer necessary in Python >=3.0 (redundant-u-string-prefix)
orig.py:88:0: C0116: Missing function or method docstring (missing-function-docstring)
orig.py:1:0: W0611: Unused import os (unused-import)
orig.py:1:0: W0611: Unused import sys (unused-import)

-----------------------------------
Your code has been rated at 4.43/10
```

AFTER PYLINT

```
PS C:\Users\admin\OneDrive\Desktop\sep_lab\L5_63_subscription-billing> py -m pylint billing_system.py

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

PS C:\Users\admin\OneDrive\Desktop\sep_lab\L5_63_subscription-billing>
```

BEFORE BANDIT

```
[main]   INFO  profile include tests: None
[main]   INFO  profile exclude tests: None
[main]   INFO  cli include tests: None
[main]   INFO  cli exclude tests: None
[main]   INFO  running on Python 3.13.16
Run started:2026-10-16 18:27:13.731978+00:00

Test results:
>> Issue: [B413:blacklist] The pyCrypto library and its module AES are no longer actively maintained and have be
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b413-import-pycrypto
   Location: ./orig.py:8:0
7  from mako.template import Template
8  from Crypto.Cipher import AES
9

--------------------------------------------------
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'whsec_live_51HxQ2bK'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: ./orig.py:14:17
13 MAX_ATTEMPTS = 3
14 WEBHOOK_SECRET = "whsec_live_51HxQ2bK"
15 CARD_KEY = b"billing-key-1234"

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purpose
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: ./orig.py:67:29
66 def couponCode():
67     return "SAVE-" + "".join(random.choice("ABCDEFGHJKLMNPQRSTUVWXYZ23456789") for _ in range(6))
68
```

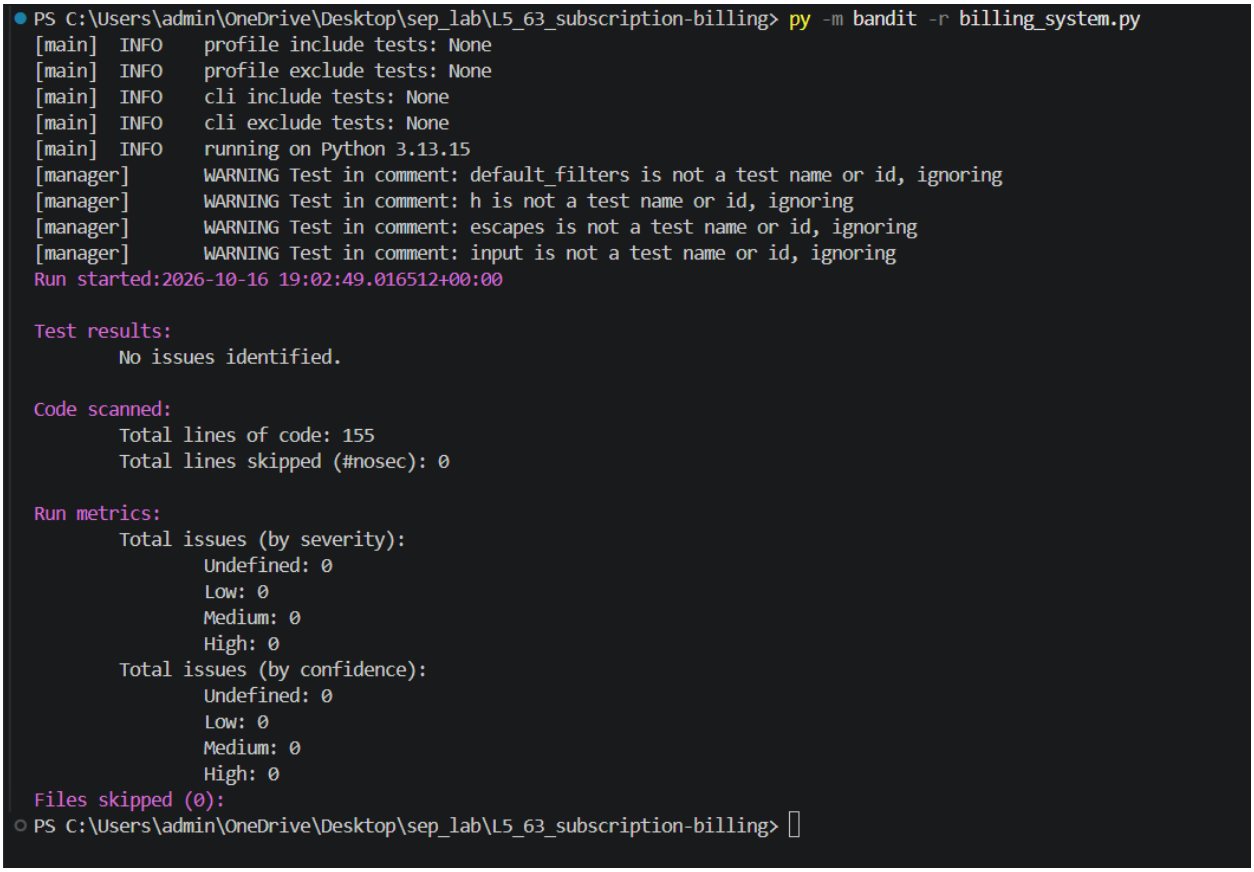
```
--------------------------------------------------
>> Issue: [B702:use_of_mako_templates] Mako templates allow HTML/JS rendering by default and are inherently
   Severity: Medium   Confidence: High
   CWE: CWE-80 (https://cwe.mitre.org/data/definitions/80.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b702_use_of_mako_templates.html
   Location: ./orig.py:86:11
85     assert amount >= 0
86     return Template(u"<h1>Invoice for ${user}</h1><p>Total: Rs.${amount}</p>").render(user=user, amount=
87

--------------------------------------------------

Code scanned:
   Total lines of code: 90
   Total lines skipped (#nosec): 0
   Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
   Total issues (by severity):
      Undefined: 0
      Low: 3
      Medium: 1
      High: 1
   Total issues (by confidence):
      Undefined: 0
      Low: 0
      Medium: 1
      High: 4
Files skipped (0):
```

AFTER BANDIT



BEFORE FLAKE

```
orig.py:1:1: F401 'os' imported but unused
orig.py:1:1: F401 'sys' imported but unused
orig.py:1:10: E401 multiple imports on one line
orig.py:20:1: E302 expected 2 blank lines, found 1
orig.py:21:80: E501 line too long (95 > 79 characters)
orig.py:24:1: E302 expected 2 blank lines, found 1
orig.py:27:1: E302 expected 2 blank lines, found 1
orig.py:29:15: E711 comparison to None should be 'if cond is None:'
orig.py:35:1: E302 expected 2 blank lines, found 1
orig.py:40:1: E302 expected 2 blank lines, found 1
orig.py:43:1: E302 expected 2 blank lines, found 1
orig.py:50:1: E302 expected 2 blank lines, found 1
orig.py:58:1: E302 expected 2 blank lines, found 1
orig.py:66:1: E302 expected 2 blank lines, found 1
orig.py:67:80: E501 line too long (97 > 79 characters)
orig.py:69:1: E302 expected 2 blank lines, found 1
orig.py:70:80: E501 line too long (80 > 79 characters)
orig.py:72:1: E302 expected 2 blank lines, found 1
orig.py:76:1: E302 expected 2 blank lines, found 1
orig.py:77:80: E501 line too long (82 > 79 characters)
orig.py:80:1: E302 expected 2 blank lines, found 1
orig.py:84:1: E302 expected 2 blank lines, found 1
orig.py:86:80: E501 line too long (111 > 79 characters)
orig.py:88:1: E302 expected 2 blank lines, found 1
orig.py:91:80: E501 line too long (94 > 79 characters)
orig.py:92:80: E501 line too long (95 > 79 characters)
orig.py:93:80: E501 line too long (97 > 79 characters)
orig.py:94:80: E501 line too long (94 > 79 characters)
orig.py:95:80: E501 line too long (80 > 79 characters)
orig.py:96:80: E501 line too long (126 > 79 characters)
orig.py:98:80: E501 line too long (103 > 79 characters)
orig.py:100:80: E501 line too long (108 > 79 characters)
orig.py:101:80: E501 line too long (80 > 79 characters)
orig.py:103:80: E501 line too long (89 > 79 characters)
orig.py:104:80: E501 line too long (82 > 79 characters)
orig.py:107:1: E305 expected 2 blank lines after class or function definition, found 1
```

AFTER FLAKE

```
PS C:\Users\admin\OneDrive\Desktop\sep_lab\L5_63_subscription-billing> py -m flake8 billing_system.py
PS C:\Users\admin\OneDrive\Desktop\sep_lab\L5_63_subscription-billing>
```